

UNIT - III

APPROACHES OF TEACHING

3.1 Steps in planning a lesson

Education should be goal oriented and objective based, otherwise it is like navigating without a magnetic compass to show and guide the direction. A key aspect of effective teaching is having a plan for what you are going to teach in the classroom. Creating such a plan involves setting realistic goals deciding how to incorporate course, text books and resource materials and develop activities that will promote learning. The following points will help in planning a lesson.

Setting lesson goals

- Building up their personality and character so that they can take their rightful place in the mainstream of the nation.
- Assimilating vocationally and socially desirable knowledge and skills.
- Developing ability to think critically and creatively.
- Developing healthy scientific attitudes towards problems and people and
- Motivating them to act according to worthy ideals in the service of community and the nation.
- Goals are based on the following order
 - ✓ Country
 - ✓ Education System

- ✓ Educational Institution
- ✓ Class
- ✓ Unit
- ✓ Lesson

All goals must be SMART

S – Specific Can the goal precise and well defined?

Is it clear?

Can every one understand it?

M – Measurable What evidence is needed to confirm the completion?

How you will judge whether the goal have been attained or not?

A – Achievable Is it within your capacity to achieve? Are there sufficient resources are available to achieve?

R – Realistic It is possible for you to perform the goal. How does it fit in to your work?

T – Timed Is there a deadline?

Is it feasible to attain in time?

Example

1. At the end of the lesson students will be able to know about how to use the numbers systems.
2. At the end of the lesson students will write c programming using valid c statements.

3. At the end of the lesson students will be able to communicate with the internet.
4. At the end of the lesson students will be able to understand the operating system.

3.2. Designing a Lesson Plan

Lesson Plan

“Classrooms determine the India’s future”. Taking the importance of the above statement, lesson plan helps the teacher to plan and prepare classroom teaching. If a teacher goes to class without any preparation, he will not only waste the time but also face some problems. Hence to plan and prepare what to teach in each period, how to present it with systematic order and evaluation procedures to be done in that class room-all combining and presenting in short is called lesson plan.

Lesson Plan Meaning

In lesson plan teacher has to clearly indicate, what are the teaching objectives and how he is going to teach the content in a specific teaching period. What are the aids to be used, what are the steps to use formative evaluation and what are the assessment are to be given etc. should be planned and noted in lesson plan.

Definitions of lesson plan

Good defines a lesson plan as an “outline of the important points of a lesson arranged in order in which they are to be presented to students by teacher”.

Bossing defines a lesson plan as an “organized statement of general and specific goals together with the specific means by which these goals are to be attained by

the learner under the guidance of the teacher on the given day”.

Standards, Lester B defines “a lesson plan is actually a plan of action. It includes the working philosophy of the teacher, her knowledge of philosophy, her information about the understanding of their pupils, their comprehension of the objectives of education, their knowledge of the material to be taught and their ability to utilize effective methods”.

Principles for preparing the lesson plan

While preparing a lesson plan, the following principles should be kept in mind.

1. Student's previous knowledge, their capabilities must be basic for a lesson plan.
2. Present the subject matter in an organized, systematic and effective manner.
3. All objectives of lesson both general and specific should know to the teacher.
4. Lesson plan should create interest in students.
5. Questions motivating self-thinking, thought-provoking etc. should find a place in lesson plan.

Need and Important features of a lesson plan

A good lesson have the following important characteristics.

1. All the objectives of the lesson both general and specific should be stated clearly in the lesson plan.
2. A good lesson plan should outline in detail the various steps that the teacher proposes to take in the class.

3. It prevents waste of time, as every step has been planned and written.
4. Lesson plan gives self confidence to the teacher.
5. A good lesson plan should indicate the list of AV aids to be used and should specify when and how to use them while teaching.
6. Sometimes we may give more time to unimportant subject matter. We can rectify this defect while preparing lesson plan.
7. A good lesson plan will integrate previous knowledge and experience to new knowledge.
8. A good lesson plan should stimulate independent thinking, interest, creative, imagination powers of the student.
9. A good lesson plan should be flexible and not rigid.
10. To complete the syllabus in time, lesson plan is needed.

Uses of Lesson Plan

1. Lesson plan helps the students to focus on the subject matter.
2. It helps to complete the lesson in time.
3. It prevents waste of time.
4. We can determine which Instructional Objectives have been fulfilled with the help of the lesson plan.
5. To become master in subject matter.
6. To get clarity in teaching activities.
7. To obtain teaching resources
8. While teaching it helps to avoid concept contradictions
9. It helps to prepare appropriate teaching aids
10. It brings self-confidence and self-power to teachers.

3.2.1 HERBARTIAN STEPS

German educator J.F. Herbart advocated five formal steps in teaching. They are

1. Preparation
2. Presentation
3. Association
4. Generalisation
5. Application

1. Introduction / Preparation

The introductory phase is called the 'warm-up' phase. The purpose is to create interest in students in the topic. The teacher may relate the new information to the one already taught or to the previous knowledge possesses by them. The new information is anchored with the existing knowledge.

2. Presentation

The transaction idea is done step by step following psychological approach. Starting from known things, the teacher leads the students to the new information by using comparisons, analogies, illustrations, demonstrations and other audio-visual aids including the chalk board.

3. Association

Through assimilation of the new idea through comparison with former ideas and consideration of their similarities and differences in order to in plant the idea in the mind.

4. Generalization

In science lessons generally the learning material leads to certain generalization leading to establishment of certain

formulae, principles or laws. An effort should be made that the students draw the conclusions themselves. Teacher should guide the students only if their generalization is either incomplete or irrelevant.

5. Application

In this step the knowledge gained is applied to certain situations. This step is in conformity with the general desire of the students to make use of generalization in order to see for them where the generalizations are valid in some other situations.

3.2.2. Preparation of lesson plan

While preparing a lesson plan generally the following steps are followed:

1. Motivation / Preparation
2. Presentation
3. Association / Comparison
4. Generalisation
5. Application
6. Recapitulation
7. Assignment

1. Motivation / Preparation

Students are encourage and motivate to learn new knowledge. The mind of the child should be prepared to receive new knowledge. Preparation may involve.

- Testing previous knowledge to learn new knowledge.

- Integrating the previous knowledge with the new knowledge.
- Arrest attention of the students using new techniques.
- Announcing the aim of the lesson.

2. Presentation:

Here teaching takes important place. Students learn new knowledge. For an effective learning outcome, teacher should ensure active student participation with the use of audio-visual aids. Teacher should stimulate thinking by asking number of questions while teaching.

3. Association / Comparison

Here the student understand the nature of the problem and compare the given information and analyse for the solution of the problem. Here knowledge is correlated with existing knowledge.

4. Generalization

In Mathematics lessons generally the learning material leads to certain generalization leading to establishment of certain formulae, principles or laws. An effort should be made that the students draw the conclusions themselves. Teacher should guide the students only if their generalization is either incomplete or irrelevant.

5. Application:

Here students apply the rules, formulas that they have learnt to solve the problems. By application of knowledge, the validity of generalization is tested and verified. Thus the knowledge becomes more meaningful to the learner and becomes permanent in his mind.

6. Recapitulation

Recapitulation helps the students to review the subject matter and consolidate ideas. Here Teacher will ascertain how well the students understood the new knowledge by way of asking few questions. From the answers we will also judge the effectiveness of the Teaching method.

7. Assignment / Home work

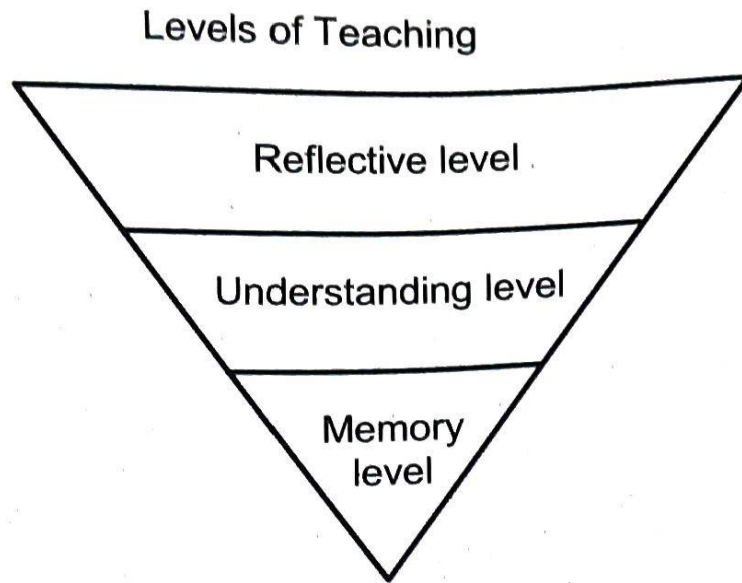
To learn mathematics effectively, class room activity alone is not enough. To know the mathematical concepts, laws, formulae and applications clearly we need home work along with class room activities. Home work revise the portions done in the classroom. Assignments, motivate and reinforce further learning.

3.3 Organising Teaching

Organising of teaching-learning is the framework on which effective teaching is based. Careful and thoughtful planning allows instructional time to be maximised, standard to be addressed, prior knowledge to be activated, misconceptions to be controlled and the diverse characteristics and learning needs to be considered. Effective teachers use different methods for teaching to teach knowledge. In this topic we are going to see three different methods for teaching.

3.3.1 Memory level of Teaching (Herbatian Model)

Teaching is a part of teaching-learning process. It is required to bring specific changes in a person according to the need of his society. It should improve the learning skills of students. Teaching has to bring changes in the behaviour of students.



Memory Levels of Teaching

Objectives

1. The objective of the memory level teaching is just to impart information or knowledge to the learners.
2. Retention and reproduction through recall or recognition.

In this level of teaching content, presentation and its exercise is important for learning. Emphasis is given on the presentation of information and facts in a systematic manner with a view to develop memory of students. At this level recall and recognition has important place.

Professor Herbart described the following five steps for the memory level of teaching. They are

- i. Focus
- ii. Syntax
- iii. Social system
- iv. Support system
- v. Evaluation system

We will discuss the above steps in detail.

i. Focus

Here importance is given on cramming of facts and development of following capacities.

- Training of mental aspects
- Providing knowledge of facts
- Retaining the learnt facts
- Recalling and Representing the learnt content

ii. Syntax

Herbert's five steps are

1. Preparation :

Important function of this step is to bring relevant ideas of the subject matter. The teacher plans about presentation of the contents.

2. Presentation :

The students are given new knowledge by testing their previous knowledge.

3. Comparison :

In this level teacher attempts to compare old facts and new facts, its similarities and dissimilarities, relationship between two subjects etc.

1. Generalization :

Pupils are given opportunities to think the content. Pupils formulate some principles and laws which can be used in the future life situations.

2. Application :

Teacher generate new situations to use learnt knowledge in new situation.

iii. Support System

Here supportive devices are used to make learning more effective. The Audio visual aids are used to present the content in an observable form.

iv. Social System

This system consists of authoritarian behaviour. The teacher is more active to construct and present the subject matter systematically and logically in the classroom. The learners are passive listeners and they have to memorize and reproduce the content and their activities are directed by the teacher. At this level teacher provides motivation through verbal praise or punishment. If the learner is motivated or reinforced continuously, then they learn better.

v. Evaluation system

Evaluation is the evidences about the realization of goals of teaching. Usually oral tests are given to measure the learnt material. The objective tests of recall and recognition types are given to evaluate the knowledge of the student.

Features of Memory Level

- Analysis of the subject matter is more important in this level of teaching. Much emphasis is given on the presentation of subject matter and not the experience of the students.
- Definite and systematic orders of subject matter is presented in this level of teaching.
- Teachers are not very flexible in the presentation of subject matter.
- The Mechanical handling of the various segments of content.

- Learning outcomes is of short duration.
- Interaction between teacher and students is at the lowest level.

Role of Teacher

- Dominating, active and authoritarian role
- He is the authority to instruct, deliver, direct, control and evaluate the performance.
- Takes initiatives, present the subject matter, directs the learner to do the activity in the desired manner.
- Right from the selection of the subject manner to its evaluation, the key process of teaching-learning remains with the teacher.

Suggestions for Memory Level Teaching

- Frequent recall or reproduction of content should be practised.
- Whole to part method should be used
- The content should have definite structure
- Teaching content should be integrated
- Contents should be presented systematically.
- Continuous reinforcement should be provided to the students.
- Students retention can be increased by practice.

Merits of Memory Level Teaching

1. Useful for children at lower classes.
2. The role of the teacher is important in this level of teaching and he is free to make choices of subject matter, plan it and can present it at will.

3. The knowledge acquired at memory level teaching forms a basis for the future.
4. Memory level teaching acts as the first step for understanding and reflective levels of teaching. It is pre-requisite for understanding level teaching.
5. It is useful for slow learners

Demerits of Memory Level Teaching

1. This does not contribute to the development of the students' capabilities.
2. At this level students learn by rote, so the knowledge gained does not prove helpful in real life situations as it does not develop the talents of students.
3. The pupils are kept in strict discipline.
4. Intelligence does not carry any importance in this type of teaching and it lacks motivation.
5. No interaction in the class
6. There is no guarantee of good retention and appropriate reproduction of the memorized material by the learner.

3.3.2 Understanding Level of Teaching (Morrison Model)

The teaching at the understanding level is of a higher quality than the one of the memory level. It is more useful and thoughtful from the point of view of mental capabilities. At this level of teaching, the teacher explains the student about the relationship between principles and facts and teach them how the principles can be applied. This method enables students to have complete command over subject material. No Cramming is encouraged here. The term understanding means to comprehend, to grasp, and to have knowledge of, to learn, to interpret and to infer etc.

Henry C Morrison clearly stated that understanding is not merely being able to recall something. It is not mere generalization deducted from specific facts; it is an insight into how it may be used in future situations. Morrison asserted that the outcome of all teaching is 'Mastery' and not memorization of facts. Morrison is the main promoter of understanding level promoter of understanding level of teaching.

Objectives :

1. Acquisition of required knowledge / facts
2. To see the relationship between acquired facts

Morrison has described five steps in his teaching model. They are

1. Exploration
2. Presentation
3. Assimilation
4. Organisation
5. Recitation

Let us discuss the above steps in detail

1. Exploration

In this step, the teacher tries to know the students 'entry-level behaviour' (Previous knowledge)

2. Presentation

The overview of the subject matter is given and the structure of the whole unit is discussed with the students.

3. Assimilation

In this step, the students study the subject matter deeply and try to relate presented material to their internal self.

4. Organisation

The student present the acquired knowledge in a systematic manner without the help of teacher.

5. Recitation

This step is related to the students' verbal expression of the subject matter which is learned by them. They visit the whole content.

Morrison's Instructional Pattern

Morrison's teaching plan involves the following steps

1. Pretest
2. Teaching
3. Testing the result of instruction.
4. Changing the instruction procedures and
5. Teaching and testing again until the unit has been completely mastered by the student.

Subject matter and its presentation

The subject matter for the understanding level teaching is quite structured in terms of planning sequential and organised presentation and meaningful learning. It needs to be linked with the previous knowledge of the students and with its utilization for acquiring new facts and application to practical life.

Role of the teacher

Like memory level of teaching, the teacher plays a dominant and authoritarian role at the understanding level of teaching. This type of teaching is also subject centred and hence teacher has to pay his full attention in presenting the subject matter to his students to attain the desired understanding objectives besides the knowledge objective.

He has to take care of the fact that the students gain complete mastery over the subject matter in term of its full understanding and generalized insight rather than its mere memorization. Lecture, Demonstration or explanation, question and answer method, inductive followed by deductive approach and analytic method with synthetic approach are all carried out at the understanding level.

Evaluation

Well-planned oral written and practical examinations are conducted in and outside the classroom. However emphasis is given to test to know ability to analyse, synthesize and apply the generalised principles for deducing new facts and solving novel problems.

Merits of Understanding Level of Teaching

1. At this level of teaching students to make use of their thinking abilities.
2. Knowledge acquired at this level forms the basis of the reflective level of teaching
3. Here the teacher present subject matter before the students in an organized and sequential form. The new knowledge acquired is related to the previously acquired knowledge.
4. Here the students do not learn by rote. They learn by understanding the facts and information and their use and purpose.

Demerits of Understanding Level of Teaching

1. Teaching at this level is subject centred. There is no interaction between the teacher and students at this level.
2. In this type of teaching mastery is emphasized.
3. Ignore higher cognitive abilities.

3.3.3 Reflective Level of Teaching (Bigge and Hunt Model)

This level of teaching is the highest level of teaching - learning activity. Reflective level of teaching means, problem centred teaching. Hunt is the main promoter of reflective level of teaching. Reflective level teaching does not depend upon the memorization, understanding of concepts, and their application only; it demands the use of higher mental processes as

- Reasoning
- Thinking
- Analysing concepts
- Imagination
- ideas and thoughts

The learner will develop new insight to

- solve problems
- to discover
- to present generalization in the light of the new evidence
- to explore the field of knowledge
- to test and retest the achieved facts.
- This level basically involves the use of the scientific method.

Objectives

1. To develop insight into solve problems
2. To develop rational and critical thinking
3. To develop the ability of independent thinking and decision making.

Model of Reflective level of Teaching

Hunt developed the following steps for the Reflective level of Teaching. They are

a. Focus

To develop problem solving, critical and constructive, independent, original thinking

b. Syntax

Steps

- i. Creating a problematic situation
- ii. Formulation of Hypothesis
- iii. Verify hypothesis
- iv. Collection of data
- v. Testing of hypothesis

c. Social system

Student occupies the primary place and teacher secondary place.

d. Support system

Objective type test is not used but essay type test is used. Attitude, beliefs and involvement is evaluated.

Features of Reflective level of teaching

- It is the highest level of teaching.
- It is problem - centric approach of teaching.
- The students are assumed to adopt some sort of research approach to solve the problem
- Classroom environment is independent. The learners are self-motivated and active.
- It leads to the development of creative skills.

- The students can solve problems of their lives by reasoning logic and imagination and lead successful and happy lives.
- Essay type test is used for evaluation. Attitude, belief and involvement are also evaluated.
- clear goals for pupil learning

Methods used for Reflective level of teaching

- Problem-solving method
- Investigating projects
- Heuristic method
- Experimental method
- Inquiry oriented method
- Analytic method

Merits of reflective level teaching

1. The teaching at this level is not teacher - centred or subject-centred. It is learner-centred.
2. There is an interaction between the teacher and the student at the reflective level teaching.
3. At this level, teaching is appropriate for the higher class.
4. At this level, teaching is highly thoughtful and useful than the teaching at the memory or understanding level.
5. This method is very useful for gifted children.

Demerits of reflective level teaching

1. It is not suitable for small children. It is suitable only for mentally matured children.

2. At this level, the study material is neither organised nor pre-planned. Therefore students cannot acquire systemic and organised knowledge of their courses.
3. It is a time consuming process.
4. It is not suitable for Indian class room.
5. It is not applicable for dull students.

3.4. Designing a unit plan

Different meaning has been assigned to the term unit. They are i) The lesson of the day ii) As a chapter in a textbook etc. syllabus contains many units such as Introduction of computers, Number Systems, Operating System, Computer Communications, Unix, Windows Explorer, Windows XP etc.

Definition of unit

Preston defines a unit as, "large block of related subject matter as can be over viewed by a learner."

Samford defines as, "A unit is an outline of carefully selected subject matter which has been isolated because of its relationship to pupil's need and interests."

Characteristics of a Good unit

- A unit should be meaningful segments of well-organized subject matter.
- General and specific instructional objectives should be clearly defined in the unit plan.
- A unit should consist of familiar and related subject.
- It should give new experience and knowledge to students.
- It should help the individual differences in learning.
- A unit should not be too lengthily or too short.

- The length of the unit should be such as to retain the interest of the students.
- It should give new experience to the students.

Unit planning

A unit plan involves planning of teaching a unit, teaching methods, evaluation of teaching activities, diagnosing and remedial steps all together is called unit planning.

Proforma of a unit plan

Name of the Teacher :

Subject : Computer Science

Standard :

Unit :

Class and section :

Total no. of periods :

Sl. No.	Subject – Units	Content analysis	Objectives and specifications	Teaching and learning activities	Teaching aids used	Evaluation

Advantages of unit planning

1. Since we are dividing the unit into small parts and teach to the students, the students are able to understand the subject matter clearly.
2. Unit plan helps the teacher to teach the students in systematic manner.

3. Unit plan help the students to understand the relationship between the facts and principles that make up the unit.
4. Unit plan help us to study all information pertaining the unit.
5. Unit plan helps to teach according to individual difference of students.
6. It helps the teachers to prepare teaching aids.
7. It helps to develop self confidence in students.

Steps followed to perform unit plan

The following steps have to be followed for a unit teaching - learning and activities.

1. Analysis of content
2. General instructional objectives
3. Specific instructional objectives
4. Planning the learning activities
5. Evaluation procedure
6. Remedial measures

1. Analysis of content

After selecting a unit, teacher has to analyse the contents of the unit in-depth. This helps the teacher to break up the unit into meaningful sub-units and retain the continuity throughout the unit.

2. General instructional objectives

Knowledge of content skill learnt is analysed her.

3. Specific instructional objectives

Specific instruction objectives of the content namely differentiating, establishing relationships, giving examples comparing, interpreting etc. are analysed.

4. Planning the learning activities

This part deals with the statements activities in teaching learning process keeping in mind individual difference, the psychology of learning and objectives. The teacher also has to plan suitable teaching methods that are to be employed here.

5. Evaluation procedure

Teacher has to evaluate with reference with reference to aims at the end of the unit or sub unit. Achievement test may be done and students' progress and the effectiveness of teaching strategies may evaluate.

6. Remedial measures

After analyzing the achievement test of the students, take remedial measures if the performance is not satisfactory.

Method of Performing Unit Plan

Computer Science syllabus for the class may be divided into many units. Each unit is to be divided into small parts/lessons and then decide the number of periods to be allotted for each lesson. Plan all the units of the syllabus has to be completed in time. According to content of the unit divide it into required number of small parts require number and allot of periods

Format to divide the syllabus into units

Class: XI

Unit No.	Name of the unit	Topics	No. of Periods required	Remarks
1	Fundamentals of Computer and working with a typical operating system	1. Introduction to computers 2. Number system 3. Computer organization 4. Theoretical concepts of operating system 5. Working with typical operating system Part I – Working with windows Part II – Working with Linux	26	Achievement test will be conducted.
2	Algorithmic Problem Solving	1. Specification and abstraction 2. Composition and decomposition 3. Iterational and recursion	20	Conducting achievement test

Signature of the Teacher

After dividing the syllabus into units and into small topic, prepare the teaching plan for each lesson/topic and complete it as per the time allotted to it.

Content of a unit – Introduction to Computers

Sl. No.	Sub Division of the content	No. of periods required	Teaching Method	Resource Materials	Evaluation
1.	Introduction to Computers	1	Demonstration method	Computer	Who is the father of computer.
2.	Generation of Computers Data and information	1	Demonstration method	Computer	Discuss various generation of Computers
3.	Components of a computer	1	Demonstration method	Computer	Write the components of a computer
4.	Different types of mouse Printers	1	Demonstration method	Different kinds of mouse	Differentiate optical and laser mouse.
5.	Revision and Test	1			Achievement test will be conducted

3.4.1 Difference between lesson plan and unit plan

S. No.	Lesson plan	Unit plan
1	Planned for a small portion to give effective teaching in one period.	Planned to teach a large content continuously for 6 to 10 periods or more.
2	Lesson plan is partial.	Unit plan is complete.
3	Full concept in learning is not achieved.	Full concept in learning is achieved.
4	Teaching alone will take place.	Teaching, evaluation, remedial teaching will take place.
5	Questions will find a place in lesson plan to motivate self-thinking, language ability etc.	There are not questions in unit plan.

6	Assignments will find a place at the end.	There are no assignments.
7	Lesson plan is written for each period.	According to length of the unit, unit plan is written for 6 to 10 periods or more.
8	Method of teaching is not specified here.	Method of teaching is specified here.

3.5 Structure of a four-fold lesson plan

Introduction

We use the introduction to tell us a little about our lesson plan. Briefly describe the instructional techniques, what students to learn and any activities or assessments that you think are particularly noteworthy. The title of the lesson plan should be concise, clear and descriptive. It should invite teachers to take a closer look at the plan. Remember that teachers may see only the title and a short abstract of our plan in a page of search results.

Objectives

We should give some general and specific objectives of the teaching lesson. These objectives should be written clearly to achieve the learning outcomes. Learning outcomes are what students are expected to learn after completing the lesson plan.

Learning outcomes should be closely related to the curriculum alignment but should not simply repeat goals and objectives of the standard course of study. Learning outcomes may be broader, address particular aspects of curriculum objectives or teach the curriculum in a special context. Each learning outcome should be clearly reflected in the activities and assessed at the conclusion of the lesson.

Session time

Session time is the amount of time a teacher will need to schedule for the lesson plan. We might specify minutes, hours, class periods.

Materials

Materials needed include resources used by both teacher and student, including books, handouts, teaching aids, paper and pencils, art supplies and so on.

Previous knowledge of Learners

Write the previous knowledge required to understand the lesson plan.

Procedures

We should follow the following procedures to take the lesson.

1. Motivation
2. Preparation
3. Presentation
4. Review/Recapitulation
5. Follow up/Assignments

1. Motivation

- It could be through interesting introduction.
- The essence of introduction consists of arousing the interest of the pupil by supplying a motive for the study of a new unit.
- It is therefore better to call motivation.
- The essence of motivation consists in making the pupil understand what they are about and feel the worth and wellness of studying it.

The most desirable motivation is for the pupil to see purpose in what he is doing. He must

- Recognize their need and its significance
- Identify the cause of his difficulties
- Have confidence in his ability to correct the difficulties

4

5.

- Be given the proper exercise materials, which provide self-evaluation, and
- Be provided with opportunities for application of skills they have learnt

2. Preparation

- This is for laying the basic foundation for construction of the new lesson.
- Test the previous knowledge of the students, so that they could understand the present topic based on those previous knowledge/concepts.
- If the pre requisite concepts are not clear, they may not understand the present concept.

3. Presentation

- The most important part of teaching all skills learnt through micro teaching must be applied here to make the teaching effective.
- Make thorough content analysis-concept by concepts and then present the concept from simple to complex, from easy to difficult etc.
- Provide various activities group work, demonstration, explanation, questioning and blackboard work.

4. Review

- Provide the summary of concepts taught in that period.
- It may be fast restating by teacher or by asking questions and eliciting answers from students.
- This also will help to identify which part not understood well by students and brief explanation could be given.

5. Follow up

- It could be or simple assignment of asking to solve problems in exercise or
- Project work.

Students may be asked to observe from society or computer centers. It could be referring to nets/webs books etc.

The structure of four-fold lesson plan is given below.

Date : xxxxx
 Name of the Student-teacher : xxxxx
 Subject : xxxxx
 Class/Section and Session : xxxxx
 Unit : xxxxx
 Name of the School : xxxxx
 Topic : xxxxx

Instructional objectives:

Write the list of objectives of the topic to be covered. i.e. Write what the student knows at lesson completion.

Instructional Resources Required:

Write the list of materials/aids required to explain the topic.

Previous Knowledge of learners:

Write the previous knowledge required to understand the new unit.

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Preparing and giving motivation to receive new knowledge / sequence of events that make up the	Behavioural objectives are the smaller, observable, understandable and measurable. It must describe	Learning experience refers to all interaction and practice in which learning takes place in classroom. In this column teacher /	On basis content concept and learning experience formative

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Lesson / Independent practice - Recapitula- tion	learning outcomes.	student activities will take place to attain the objectives.	questions are framed here step by step.

Followup activities : Give homework and assignment in the topic covered.

Signature of Guide Teacher

Signature of the Student Teacher

3.6 Preparation of a model lesson plan

Model Lesson Plan - 1

Date : xxxxx
 Name of the Student-teacher: xxxxx
 Subject : Computer Science
 Class/Section and Session : XI A
 Unit : C Programming
 Name of the School : xxxxx
 Topic : Unary Operators

Instructional objectives:

The pupil,

1. Knows about computer symbols.
2. Understands about operators.
3. Understands about arithmetic operators.
4. Acquires knowledge about unary operator.
5. Explains about operators.
6. Recalls the arithmetic operator.

7. List out some examples for arithmetic operator.
8. Illustrates the unary operator.
9. Describes examples for unary operator.

Instructional Resources Required:

1. Textbook 11th std Tamilnadu Government Corporation
2. www.google.com
3. Chart
4. Flash cards
5. Round cards
6. PowerPoint Presentation

Previous Knowledge of learners:

The learners have a previous knowledge about the arithmetic and unary operator that they have learnt in C language in the previous standard.

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
1. Motivation: The motivation was given by asking questions to the students. Have you seen some traffic symbols? What are they? What are uses of it?	The students get motivated and interested towards the topic.	The students are brought to the main topic and they are ready to learn.	Which topic today we are going to discuss?
2. Introducing the lesson: The teacher introduces the lesson by writing the topic.	Students will be able to understand what topic is going to discuss?	The students observe and writes the in their note.	

[illegible]

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
must have arithmetic or pointer type and the result of the argument. Example: If $a = 5$ then $+a$ means 5 If $a = -4$ then $+a$ means -4 Unary + : The unary + operator have arithmetic type and the result is negation of its operand's value. Example: If $a = 5$ then $-a$ means -5 If $a = 0$ then $-a$ means 0	Recognizes Understands Knows	The teacher explains unary + operator. The students understand the meaning of unary operator. The teacher the teachers example. The student understands the example. The teacher explains unary -. The students know unary -. The teacher explains the example. The students knows example.	___ type? If $a = -3$ then $+a$ means? The unary operator must have ___ type? If $a = -2$ then $-a$ means?

4. Review:

1. The operations being carried out on data are represented by ___?
2. What are the arithmetic operators?

3. What is called unary operator?
4. What are the types of unary operator?
5. The operand of the unary + operator must have ____ type?
6. If $a = -3$ then $+a$ means?
7. The unary- operator must have ____ type?
8. If $a = -2$ then $-a$ means?

5. Follow up activities:

Write some examples for arithmetic and unary operator?

Signature of Guide

Signature of student Teacher

Model Lesson Plan - 2

Date	:	xxxxx
Name of the Student-teacher	:	xxxxx
Subject	:	Computer Science
Class/Section and Session	:	XI C
Unit	:	Windows XP
Name of the School	:	xxxxx
Topic	:	The Mouse

Instructional objectives:

The pupil,

1. Knows about the mouse.

2. Understands the concept of performing action through mouse.
3. Acquires knowledge about the uses of mouse clicks.
4. Applies acquired knowledge in future lessons.
5. Recalls the concept of mouse.
6. Recognizes the use of buttons in mouse.
7. Explains the operations of mouse.
8. Describe the equivalent keys for mouse operations.

Instructional Resources Required:

1. Textbook 11th std Tamilnadu Government Corporation (Higher Secondary First Year Volume-II)
2. Chart (concept of mouse)
3. Flash card (key for mouse operations)
4. Model (mouse)
5. PowerPoint Presentation

Previous Knowledge of learners:

The learners have a previous knowledge of mouse, its usage and functions which leads to the concept.

Content/Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
1. Motivation: The teacher motivated the students by asking "Have you ever seen the computer mouse"	Motivated	The teacher shows the model of a mouse and the students observes it.	

Content/Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
and the students answered to it. 2. Introducing the lesson: The teacher introduced the topic from the motivation as mouse.	Observed	The teacher writes the topic on the blackboard.	
3. Explanation: If you want to extract work from the computer, you have to input data. The input can normally be provided by the keyboard and the mouse. The mouse can be used by simple pointing to it and selecting. The mouse is an input device that you move on a flat surface (usually mouse pad) when you move the mouse, a pointer moves on the screen.	Acquires	The teacher shows the chart and the student observes it.	What do you mean by mouse?
Mouse operations: Move: Moving the mouse is simply dragging the mouse on the mouse	Recalls	The teacher explains the concept and the student take down the notes.	What is the move operation stands for?

Content/Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
pad so that mouse pointer moves in the direction you want without touching the buttons. Click: Clicking is used to select objects on the windows screen. Double Click: Double click is most often to start applications. To double click, point to what you want and press the left button of the mouse twice. Click and Drag: The mouse action is used to move an object from one place to another. When you click and drag an object, objects moves along with the mouse pointer. Mouse after right click: Right mouse button gains a lot of significance now-a-	Explains	The teacher explains and the student listens.	What do you mean by context sensitive menu?

Content/Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
days. If you right click on the item, you will be provided with the context sensitive menu (it changes its contents depending on the situation) If you want to use your keyboard to do the work of the mouse. You have to follow these steps. - Click the Start button. - Select the Control Panel in the menu and click it. - Choose the Accessibility Options icon and click on it. - It opens a screen, click on Accessibility Option under pick a Control Panel icon. - Open the Mouse tab. - Activate use MouseKeys check box if it is not already activated. The equivalent keys for mouse operations.	Recognizes	The teacher explains the concept and the student takes notes.	How to choose accessibility option?

Content/Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Key Equivalent Action 5 Click + Double Click / Select the left mouse button * Select both mouse buttons - Select the right mouse button Insert Lock the selected button Delete Release the selected button	Describes	The teacher shows the flash cards and the students recalls.	What is the equivalent key for double click operation?

4. Review:

1. What do you mean by mouse?
2. What is the 'move' operation stands for?
3. What do you mean by context sensitive menu?
4. How to choose accessibility option?
5. What is the equivalent key for the double click operation?

5. Follow up activities:

1. Draw the model of mouse in your homework note?
2. Revise and come today's class.

Signature of Guide

Signature of student Teacher

Model Lesson Plan - 3

Date : xxxxx
Name of the Student-teacher: xxxxx
Subject : Computer Science
Class/Section and Session: XI C
Unit : Computer Communication
Name of the School : xxxxx
Topic : Network Topology & Basic
Elements in Networking

Instructional objectives:

The pupil,

1. Understands the process of network topology.
2. Classify the types of network.
3. Explains the workage of star network.
4. Knows how the ring network works.
5. Analyse the usage of bus networks.
6. Establish the process of hybrid network.
7. Describe the process of hybrid network.
8. Illustrate the basic elements in networking.

Instructional Resources Required:

1. Textbook 11th std Tamilnadu Government Corporation
(Higher Secondary First Year Volume-I)
2. Chart (types of topology)
3. Flash cards (Basic elements in networking)
4. Model (Ring Network)
5. PowerPoint Presentation

Previous Knowledge of learners:

Students have knowledge about networks and types of networks. So now they are going to study in detail with various types of network topology.

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
1. Motivation: In previous class we discussed about LAN, MAN and WAN with their working process, here the modes of data transmission differ. Same as various networks are connected together to perform the action and it is called "Network Topology"	Students answered to the question that was asked by the teacher.	Students were motivated by the teacher.	
2. Introducing the lesson: The teacher introduced the topic by writing on blackboard.	Observes	The students came to know about the topic.	
3. Explanation: Network Topology It is the structure or layout of the communication	Under-stands	The teacher explained by showing the	What is network topology?

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
channels that connects the various computers on the network. There are number of factors that determine the topology. Some of the important consideration is the type of nodes, the expected performance type of wiring. Types of Network Network can be laid out in different ways. The five common topologies are 1. Star 2. Ring 3. Bus 4. Hybrid / Mesh 5. FDDI These are the five common topologies.	Classifies	chart and students were observing. The teacher was explaining and students were taking notes.	Common topologies are _____?

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
1. Star Network In a star network all computers and other communication devices are connected to a central hub. Such as a file server or host computer usually by a unshielded twisted pair (UTP) cables.	Explains	The teacher was teaching and students were listening.	Abbreviate UTP.
2. Ring Network In a ring network computers and other communication devices are connected in a continuous loop. Electronic data are passed around the ring in one direction with each nodes serving as a repeater until it reaches the right destination. There is no central host computer or server.	Knows	Students asked doubt with the teacher and teacher clarified their doubt.	What is ring network?

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
3. Bus Network All communication devices are connected to a common cable called bus. There is no central computer or server. The data transmission is bidirectional.	Analyse	Teacher explained and students were observing.	Explain the process of bus network?
4. Hybrid Network A hybrid network is a combination of the above three networks.	Establishes	Students were observing the class.	Hybrid network is combination of _____?
5. FDDI Network FDDI (Fiber Distributed Data Interface) is a high-speed network using fibre optic cable. It is used for high tech purpose such as electronic images, high resolution graphics and digital video. The main disadvantage is high cost.	Describes	Teacher asked the question and students replied.	Abbreviate FDDI.

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
Basic elements in Networking All networks require the following three elements. 1. Network service 2. Transmission Media 3. Protocols	Observes	Teacher explains by showing flashcards.	Three basic elements are ____.
1. Network Service Network services are provided by numerous combination of computer hardware and software network services require data input/output.	Illustrate	Teacher explained by showing the flashcard and students observe.	What is transmission media?
2. Transmission Media It is the pathway for contacting each computer with other. Transmission media include cable and wireless, technologies that allow devices to contact each other.	Understands	Through flashcard concept was explained and students observe.	A protocol is a ____.

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
3. Protocols A Protocol can be one rule or a set of rules and standards that allow different devices to hold conversations.			

4. Review

Network topology is the layout of the communication channels and five common topologies are star, ring, bus, hybrid and FDDI and the basic elements in networking are network service, transmission media and protocol.

5. Follow up activities

Connect the types of network topology and write the difference between them.

Signature of Guide

Signature of student Teacher

Model Lesson Plan - 4

Name of the Student Teacher : xxxx
 Class / Section and Session : VII Std. B AN
 Name of the school : xxxx
 Subject : Mathematics
 Unit : Real Number System
 Topic : Addition of fractions

Date of teaching : XXXX

Instructional objectives

Student

1. Acquires knowledge to find the addition of fractions.
2. Understand to find the addition of fractions.
3. Knowledge and understanding of addition of fractions is applied, to new and unfamiliar situations.
4. Does calculations with speed and accuracy.
5. Verifies the result

Instructional Resources Required

1. Chart depicting fractions

Previous Knowledge of learners :

Knowledge of proper, improper and mixed fractions.

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Motivation	Recalls	A proper fraction is of the form p/q where $p < q$ A Improper fraction is the form p/q where $p > q$	What is a proper fraction? What is a improper fraction?
		A Natural number and a proper fraction, combined into one number is called mixed fraction.	What is called a mixed fraction?
		Teacher interact with students and extract answers. Students are giving examples.	
Introducing the topic	Compares	Teacher announces the topic as "Addition of fractions" and write the topic on the	

Let us add two fractions $\frac{3}{7} + \frac{4}{7}$	Presentation	Black board and explains Teacher Explains How to add two natural numbers. Then in the same way explains for fractions shows the Chars of depictions fractions and explains.	What is $\frac{3}{7} + \frac{4}{7}$
	Recalls Acquires knowledge	LCM of 7, 7 is 7 $\frac{3}{7} + \frac{4}{7} = \frac{3+4}{7} = \frac{7}{7}$ $= 1$	What is the LCM 7, 7
Let us add two fractions $\frac{3}{4} + \frac{1}{6}$	Acquired knowledge	LCM of 4, 6 is 12 $\frac{3}{4} + \frac{1}{6} = \frac{3 \times 3 + 1 \times 2}{12} = \frac{9+2}{12} = \frac{11}{12}$	What is the LCM of 4, 6
Simply $5\frac{1}{4} + 4\frac{3}{4} + 2\frac{5}{8}$	Recalls	$5\frac{1}{4} = \frac{5 \times 4 + 1}{4} = \frac{21}{4}$ $4\frac{3}{4} = \frac{4 \times 4 + 3}{4} = \frac{19}{4}$ $2\frac{5}{8} = \frac{2 \times 8 + 5}{8} = \frac{21}{8}$	What is the Improper fraction of $5\frac{1}{4}$ What is the Improper fraction of $4\frac{3}{4}$ What is the Improper fraction of $2\frac{5}{8}$
	Recalls Establish relationship	LCM of 4, 4, 8 is 8 $5\frac{1}{4} + 4\frac{3}{4} + 2\frac{5}{8}$ $= \frac{21}{4} + \frac{19}{4} + \frac{21}{8}$ $= \frac{21 \times 2 + 19 \times 2 + 21 \times 1}{8}$ $= \frac{42 + 38 + 21}{8}$ $= \frac{101}{8} = 12\frac{5}{8}$	What is the LCM of 4, 4, 8? What is the mixed fraction of

Date of teaching : xxxx

Instructional objectives

Student

1. Acquires knowledge to find the addition of fractions.
2. Understand to find the addition of fractions.
3. Knowledge and understanding of addition of fractions is applied, to new and unfamiliar situations.
4. Does calculations with speed and accuracy.
5. Verifies the result

Instructional Resources Required

1. Chart depicting fractions

Previous Knowledge of learners :

Knowledge of proper, improper and mixed fractions.

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Motivation	Recalls	A proper fraction is of the form p/q where $p < q$ A Improper fraction is the form p/q where $p > q$	What is a proper fraction? What is a improper fraction?
		A Natural number and a proper fraction, combined into one number is called mixed fraction.	What is called a mixed fraction?
		Teacher interact with students and extract answers. Students are giving examples.	
Introducing the topic	Compares	Teacher announces the topic as "Addition of fractions" and write the topic on the	

		Black board and explains	
Let us add two fractions $\frac{3}{7} + \frac{4}{7}$	Presentation	Teacher Explains How to add two natural numbers. Then in the same way explains for fractions shows the Chars of depictions fractions and explains.	What is $\frac{3}{7} + \frac{4}{7}$
	Recalls Acquires knowledge	LCM of 7, 7 is 7 $\frac{3}{7} + \frac{4}{7} = \frac{3+4}{7} = \frac{7}{7}$ $= 1$	What is the LCM 7, 7
Letus add two fractions $\frac{3}{4} + \frac{1}{6}$	Acquired knowledge	LCM of 4, 6 is 12 $\frac{3}{4} + \frac{1}{6} = \frac{3 \times 3 + 1 \times 2}{12} = \frac{9+2}{12} = \frac{11}{12}$	What is the LCM of 4, 6
Simply $5\frac{1}{4} + 4\frac{3}{4} + 2\frac{5}{8}$	Recalls	$5\frac{1}{4} = \frac{5 \times 4 + 1}{4} = \frac{21}{4}$ $4\frac{3}{4} = \frac{4 \times 4 + 3}{4} = \frac{19}{4}$ $2\frac{5}{8} = \frac{2 \times 8 + 5}{8} = \frac{21}{8}$	What is the improper fraction of $5\frac{1}{4}$ What is the improper fraction of $4\frac{3}{4}$ What is the improper fraction of $2\frac{5}{8}$
	Recalls Establish relationship	LCM of 4, 4, 8 is 8 $5\frac{1}{4} + 4\frac{3}{4} + 2\frac{5}{8}$ $= \frac{21}{4} + \frac{19}{4} + \frac{21}{8}$ $= \frac{21 \times 2 + 19 \times 2 + 21 \times 1}{8}$ $= \frac{42 + 38 + 21}{8}$ $= \frac{101}{8} = 12\frac{5}{8}$	What is the LCM of 4, 4, 8? What is the mixed fraction of

			$\frac{101}{8} ?$
Simplify $2\frac{3}{4} + 3\frac{5}{8} + 2\frac{1}{3}$	Application	Teacher ask a student to find the sum of the given problem in black board	
	Speed and accuracy	$2\frac{3}{4} + 3\frac{5}{8} + 2\frac{1}{3}$ $= \frac{11}{4} + \frac{29}{8} + \frac{7}{3}$ $= \frac{66 + 87 + 56}{24}$ $= \frac{209}{24} = 8\frac{17}{24}$	
Teacher review the important points	Recapitulation	Teacher review the important points and ask few questions.	1. What is the LCM of 5 and 4? 2. What is the mixed fraction of $\frac{165}{8}$.

Follow up Activities

Simplify

1. $\frac{3}{8} + \frac{4}{5} + \frac{1}{2}$

2. $3\frac{1}{4} + 2\frac{2}{3} + 4\frac{1}{12}$

Signature of Guide Teacher

Signature of the Student Teacher

Model Lesson Plan - 5

Name of the Student Teacher : xxxx

Class / Section and Session : VI Std. A FN

Name of the school : xxxx

Subject : Mathematics

Unit : Geometry

Topic : Angles

Date of teaching : xxxx

Instructional objectives

Student

1. Acquires knowledge of angles
2. Understand what is an angle
3. Knowledge and understanding is applied to new and unfamiliar situation.
4. Identify the type of angles
5. Identify the angles formed by hour hand and minute hand in a clock.

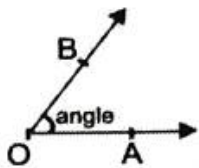
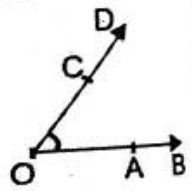
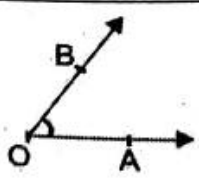
Instructional Resources Required

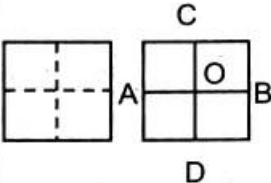
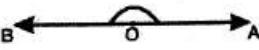
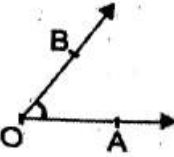
Charts depicting angle, right angle, Acute angle, Obtuse angle and Straight line angle.

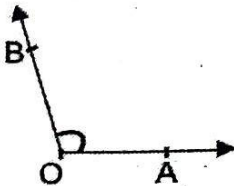
Previous Knowledge of learners :

Knowledge of point and Ray

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Motivation	Recalls	Make a point 'o' From O draw a ray $\overrightarrow{OA}$ 	What is the ray OA?
Introducing the Topic		Teacher announced the topic as, 'Angles' and write the topic on the blackboard.	

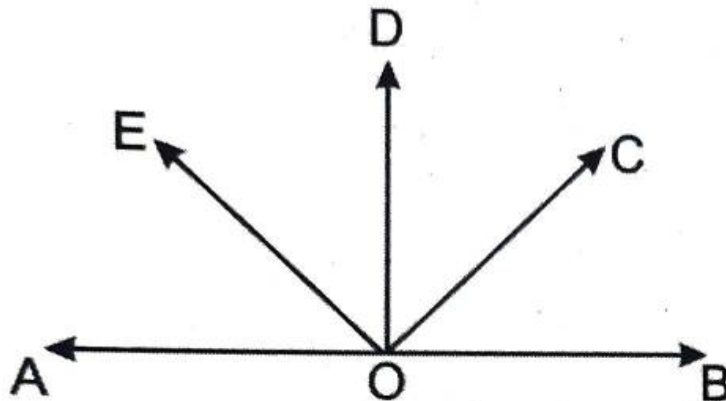
Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
	Presentation	The figure formed by two rays with the same initial point is called an angle  The angle shown in the above figure is represented by $\angle AOB$ or $\angle BOA$ or sometimes as $\angle O$	What is angle AOB
Angle	Identifies	Teacher shows a chart and ask questions $\angle AOC$ or $\angle BOC$ or $\angle BOD$ or $\angle AOC$	 Mention the angle in the above figure.
Measurer of the angle	Presentation Acquire Knowledge	 In the above figure let O as centre Ray $\vec{OA}$ rotates in the anticlock wise direction and reaches $\vec{OB}$. The rotation made by the ray is called the measure of that angle	

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Right angle	Presentation	 Fold a piece of paper as shown in the figure and unfold it. We get four angles. These angles are called right angles whose measure is 90°. Teacher explains with a chart. $\angle AOC$ or $\angle BOC$ or $\angle DOB$ or $\angle DOA$ are equal and their angle is 90°.	What is the measure of a right angle?
Straight line angle	Acquire knowledge	An angle whose measure is 180° is called a straight angle.  $\angle AOB = 180^\circ$ Teacher explains with a chart	What is the measure of a straight line angle?
Acute angle	Acquired knowledge	An angle whose measure is greater than 0 and less than 90° is called an acute angle. 	What is an acute angle?

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
		<u>AOB</u> is an acute angle Teacher explains with a chart.	
	Understands	Angle 62°, Angle 85°	Give any two examples of acute angles
Obtuse angle	Acquires knowledge	An angle whose measure is greater than 90° and less than 180° is called an obtuse angle.  <u>AOB</u> is obtuse angle Teacher explains with a chart.	What is an obtuse angle?
	Understands	Angle 160°, Angle 95°	Give any two examples of obtuse angles.
Zero angle	Acquires knowledge	 If both the rays coincide the angle formed is 0° <u>AOB</u> = 0°	
Teacher review the important points and asks few questions	Recapitulation	1. What is the measure of right angle? 2. What is an acute angle? 3. What is an obtuse angle? 4. What is the measure of straight line angle?	

Follow up Activities

1. Give any two angles of obtuse angle.
2. Name the angles and write its kind.



Signature of Guide Teacher

Signature of the Student Teacher

Review Question

1. Define Lesson plan. Write the need and importance of Lesson Plan.
2. Explain the steps followed to prepare a lesson plan.
3. Explain in detail the Memory Level of Teaching. Write its Merits and Demerits.
4. Explain in detail the Understanding Level of Teaching. Write its Merits and Demerits.
5. Explain in detail the Reflective Level of Teaching. Write its Merits and Demerits.

a Unit Plan. What are the steps to be followed to form a unit plan. Write its advantages.

7. Write the difference between Unit plan and Lesson plan.
8. Explain in detail the structure of a four-fold lesson plan.
9. Write a model lesson plan for XI Std. Computer Science.
10. Write a model lesson plan for VIII Std. Mathematics.